HOTEL BOOKING CANCELLATION PREDICTION ANALYSIS

Model Comparison Report

From Clean Data to Machine Learning

1. Objective

The objective of this project was to develop machine learning models to predict whether a hotel booking would be cancelled (is_canceled).

Three classification models were evaluated:

- Logistic Regression
- Decision Tree
- Random Forest

The models were evaluated using Accuracy, Precision, Recall, F1 Score, and ROC AUC.

2. Model Performance

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Logistic Regression	**79.01%**	**67.45%**	**45.72%**	**54.50%**	**83.25%**
Decision Tree	**81.49%**	**70.90%**	**55.38%**	**62.19%**	**86.86%**
Random Forest	**80.48%**	**78.63%**	**39.81%**	**52.86%**	**87.46%**

3. Model Selection

The **Decision Tree** was selected as the final model.

Although Random Forest achieved the highest Precision and ROC AUC, the Decision Tree achieved the highest Accuracy, Recall, and F1 Score.

For this problem, identifying actual cancellations is important. Therefore, the higher cancellation Recall of the Decision Tree (**55.38%**) made it the more suitable model for the current business objective.

4. Overfitting Check

The Decision Tree showed relatively small differences between training and test performance:

- Accuracy Gap: **0.71 percentage points**
- F1 Score Gap: **1.56 percentage points**

These relatively small gaps indicate that substantial overfitting was not observed in the selected model.

Random Forest showed a larger F1 Score gap of **5.28 percentage points**, indicating comparatively greater potential overfitting.

5. Important Features

The three most important features identified by the selected Decision Tree were:

1. country_PRT — 0.1841
2. total_of_special_requests — 0.1615
3. lead_time — 0.1605

These features provided the strongest predictive contribution within the trained Decision Tree.

6. Conclusion

The Decision Tree provided the best overall balance of predictive performance for the current hotel cancellation prediction task.

The model achieved **81.49% test accuracy, 55.38% cancellation recall, and 62.19% F1 Score**.

The model is considered a promising **decision-support prototype**, but additional validation using future and real-world booking data is recommended before production deployment.